

Project 6

The Field Guide

Build: ● Accessible

Data: ● Easy

Cost: €5-€10

Business Case

Millions of people photograph plants, insects, and birds every day, and they all want the same two things: *what is this, and what should I do about it?* Identification applications answer the first question; almost none answer the second. Is this plant invasive in my region? Is this spider dangerous to my dog? Is this mushroom protected — or lethal? Local councils, conservation groups, and agricultural bodies all need exactly this pipeline: identify, check against regional rules, and recommend an action. It is a compact, satisfying demonstration of vision, grounded retrieval, and agentic reasoning, with a genuine safety dimension.

Project Overview

You will build a naturalist assistant. The user uploads a **photograph** of a plant, insect, bird, or fungus, and optionally states their region. The system identifies the species using a vision model, retrieves what is known about it from a curated corpus of field-guide material, checks its **regional status** — invasive, protected, toxic, or harmless — and returns an actionable, cited answer. For example: *"This is Japanese knotweed. It is a regulated invasive species in England; do not compost it, and report or dispose of it as follows..."*. If you are feeling ambitious, you could use a text to video pipeline to also create an informative 45-60 seconds video about the plant/animal/insect that was given to the system.

Key Objectives

Build the vision identification path with **honest uncertainty**: present the top three candidate species with confidence levels, never a single overconfident guess. This is your multimodal core.

Build Retrieval-Augmented Generation over species accounts and regional status lists, with a citation in every answer.

Build the agent chain: identify the species, retrieve the species information, check the regional lists, and compose the action advice. If the user has not stated their region, the agent asks — and remembers the answer for the rest of the session.

Implement the safety behaviour: for anything potentially toxic or dangerous, the answer must lead with the warning, and the system must never encourage consumption. A rule to hard-code: *never eat a mushroom on the strength of an application's identification*.

Build a clean, mobile-friendly interface — this project begs to be demonstrated on a telephone.

Core Requirements

All seven core requirements from the Overview document apply: Retrieval-Augmented Generation with cited sources, an agentic architecture with tools and memory, tracing, a multimodal component (here, photograph input), an evaluation set of at least thirty tasks with benchmarking against a baseline, a deployed web application, and a final presentation of fifteen to twenty minutes. Fine-tuning is optional and discouraged.

Estimated Cost: €5–€10

The identification interface from iNaturalist is free, the species corpus and the regional lists are free, and the corpus is small. Your only real expense is language-model usage for the agent, plus vision-model calls if you choose to compare a general vision model against the iNaturalist identifier — which is an excellent evaluation angle and still cheap at this scale.

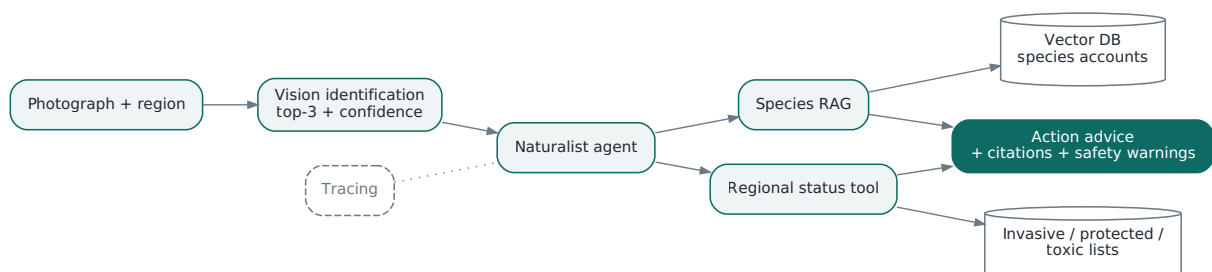
Data Plan

Everything is open, and everything is reachable through a public programming interface:

1. **Identification:** call the free **iNaturalist** computer-vision interface — or use a general vision language model, or both, and compare their accuracy.
2. **Species knowledge corpus:** species accounts from **Wikipedia**, **Wikispecies**, and **GBIF** (the Global Biodiversity Information Facility, which offers a free interface), for a scoped set of one hundred to two hundred species relevant to your chosen region.
3. **Regional status:** government-published lists — for example, the United Kingdom's invasive species lists from the GB Non-native Species Secretariat, the protected species schedules, and the Royal Horticultural Society's poisonous plants list. Choose **one region** and ingest its lists properly.

Week 1 checkpoint: the species corpus and the regional lists are indexed, and one photograph travels the full path — identification, retrieval, cited status answer.

Suggested Architecture



Suggested architecture

Timeline

Follow the shared five-week plan in the Overview document. Because the core is compact, you should reach the full flow by Week 3, leaving Weeks 4 and 5 for evaluation depth — this project supports an unusually rigorous evaluation — and interface polish.

Deliverables and Evaluation

The standard deliverables apply. For your evaluation, build a photograph test set of at least forty images with known species, and report two things: identification accuracy (both top-one and top-three) and status-check correctness — did the system correctly flag the invasive, protected, and toxic species among them? Your failure analysis should include one plausible misidentification, and discuss how the uncertainty display handled it.

Stretch Goals

- Seasonal and lookalike warnings — *"often confused with X; check the stem"*.
- A report-an-invasive flow that drafts the official reporting submission for the user's region.
- An offline-friendly mode built on a locally run vision model.